

IMST sentire™ Radar Module

24 GHz sR-1200 Series

User Manual

Version 1.5



About this Document

This document is a manual of IMST's sentire™ Radar Module. The following table displays the changes applied to this manual and assigns a version number to each document release.

Version	Date of Change	Reason of Change
1.0	2016-03-16	initial document
1.1	2016-05-02	modified specification
1.2	2016-11-15	modified Ethernet pin description
1.3	2017-03-17	modified specification
1.4	2017-11-01	added section 2.4.3 HTTP Server
1.5	2018-03-22	modified section 2.4.3 HTTP Server

Contents

1	Introduction	1
1.1	Product Description	1
1.2	Important Information	1
1.2.1	Electrical Safety	2
1.2.2	ESD Protection	2
1.2.3	RF Safety Considerations	2
1.2.4	Thermal Operating Conditions	2
1.2.5	Mechanical Safety	3
1.2.6	Chemical Conditions	3
2	Installation	5
2.1	Hardware Overview	5
2.2	SPI Interface	6
2.2.1	SPI Pin Description	6
2.2.2	SPI Parameters	7
2.2.3	SPI-to-USB Converter	7
2.3	CAN Bus Interface	8
2.3.1	CAN Pin Description	8
2.3.2	CAN Parameters	9
2.3.3	CAN-to-USB Converter	10
2.4	Ethernet Interface	10
2.4.1	Ethernet Pin Description	11
2.4.2	Ethernet Parameters	11

2.4.3	HTTP Server	13
2.5	DC Power Supply	13
3	Technical Specifications	15
3.1	Datasheet	15
3.2	Update Rate	16
3.3	Operating Temperature	16
4	Radar Fundamentals	17
4.1	Principle of Operation	17
4.2	FMCW Mode	18
4.3	Doppler Mode	20
4.4	Range-Doppler-Ambiguity	21
4.5	Angle Measurement	21
	Appendix	23
A	Referenced Documents	23
B	List of Abbreviations	23
C	List of Figures	25
D	List of Tables	26

1. Introduction

1.1 Product Description

This manual describes the IMST's sentire™ Radar Module and its installation procedure. The radar operates in FMCW or CW mode at a frequency of 24 GHz and provides the capability for range and velocity measurements in a wide area of applications. Using the integrated transmitter and receiver antennas, it is also possible to estimate the angle-of-arrival of the back-scattered signal, corresponding to a certain target direction. The radar module is available with three different interfaces (SPI, CAN or Ethernet). In case of the SPI interface, an external SPI-to-USB adapter allows to connect the radar directly to the computer. While an external 12 V power supply is required for powering the SPI or CAN bus modules, the Ethernet version uses PoE supply. The unit with Ethernet interface has CE type approval for operation in the ISM band ranging between 24 GHz to 24.25 GHz at an output power of 19 dBm EIRP (for more details please adhere to the national frequency regulations of your country). The technical specifications of the sentire™ Radar Module are summarized in chapter 3. Independently from the selected interface, the module provides an extensive set of user commands for the following tasks:

- configuration of radar parameters, as the operation mode or the frequency range,
- accessing internal radar parameters, as the current frontend temperature or the transmission power,
- triggering a measurement process and accessing the generated data,
- starting high-level applications, e.g. the Human Tracker algorithm.

However, this document doesn't contain an exhaustive explanation of all commands supported by the Radar Module. In order to access all functions of the sensor, the user can refer to the command description listed in the command interface manual /CMD/.

1.2 Important Information

The following section contains useful hints and recommendations for a save operation of the sentire™ Radar Module.

1.2.1 Electrical Safety

The sentire™ Radar Module operates at a maximum DC voltage of 54 V (PoE version, refer to chapter 3), which falls under the IEC definition of Extra-Low Voltage (ELV). Due to the non-conducting housing, no additional protection against electrical shock is required. In order to avoid damaging the sensor, it is equipped with an fuse of 0.5 A and a reverse voltage protection. Additionally, the following rules have to be followed:

- operate the device only with specified DC voltage levels,
- don't open the housing while the device is powered,
- don't intentionally change the polarity.

1.2.2 ESD Protection

The sentire™ Radar Module is protected against Electrostatic Discharge (ESD) and tested according to DIN EN 61000-4-2. While working with an open housing (e.g. connecting a new cable to the Interface board), the following rules have to be followed:

- the device must be unpowered,
- the cable must be disconnected,
- an electrical grounding is required.

1.2.3 RF Safety Considerations

The sentire™ Radar Module operates in the ISM band of 24 GHz and generates a maximum RF output power of 10 mW (10 dBm), corresponding to an EIRP power of approximately 19 dBm. With an active radome surface of 8 cm × 8 cm, the maximum RF power density is expected to be 1.56 W/m², which is significantly lower than the threshold value of 10 W/m², defined by the International Commission on Non-Ionizing Radiation Protection (ICNIRP) in 1998.

However, if several RF devices operate simultaneously in the same frequency range, different crosstalk effects can appear, leading to a reduced sensor sensitivity or erroneous measurement data. It is at the responsibility of the user to take care of these effects.

1.2.4 Thermal Operating Conditions

The housing of the sentire™ Radar Module is made of black Polycarbonate, is classified as flame-retardant and approved to the following standards:

- V-2 material class of the UL94 and IEC 60696-11-10 standards,

- B2 fire class of the DIN 4102 standard.

Additionally, the following limitations must be taken into account:

- operate the radar only in the temperature range specified in section 3,
- long term measurements at high temperatures stress the RF components and may affect the sensor's lifetime,
- internal temperatures of more than 85° temporary will deactivate the radar frontend,
- please provide additional cooling in high-temperature environments.

1.2.5 Mechanical Safety

The housing of the sentireTM Radar Module consists of two solid parts: a box containing the radome and PCB mounting structures, and a lid combined with a mounting plate. Both parts are connected with four screws. Due to a rubber sealing ring between box and lid, the radar is protected against splash water and is compliant to the IP65 standard. In order to ensure a save operation of the sensor, the following rules have to be followed:

- always use all mounting holes of the housing to fix the radar,
- don't use the attached cable for mounting reasons, don't pull the cable,
- don't mechanically stress the radar (e.g. pressure, shock, strong vibrations),
- don't cover the RF active side of the sensor (radome) with tape or other materials,
- don't use the radar if the housing is damaged.

1.2.6 Chemical Conditions

The material used for the housing of the sentireTM Radar Module is chemically resistant against dilute acids, alcohols and many greases and oils. It is stabilized against exposure to UV radiation and provides an excellent weathering resistance.

2. Installation

2.1 Hardware Overview

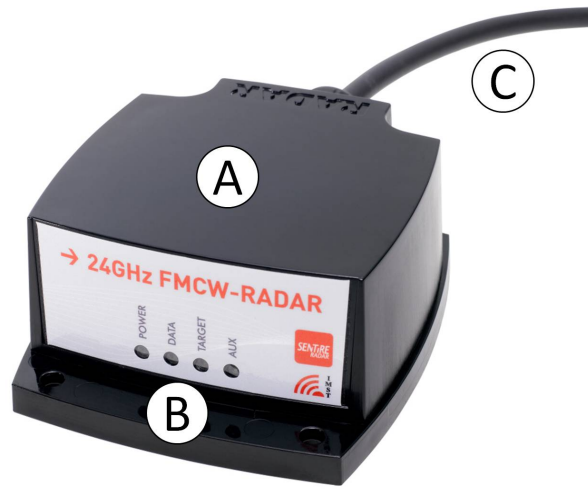


Figure 2.1: IMST sentire™ Radar Module (A: radome and antenna position, B: LED indicators, C: interface and power supply cable)

The sentire™ Radar Module (see figure 2.1) is a highly integrated device and consists of three hardware groups mounted into a robust and compact plastic housing:

- *Interface Board*, containing the power supply and the communication module (SPI, CAN or Ethernet),
- *Baseband Board*, containing the built-in digital signal controller (DSC) and a data acquisition unit,
- *Frontend Board*, containing the RF components as well as the integrated patch antennas.

Interface, Baseband and Frontend boards are stuck together using multi-polar connectors. The housing is almost completely transparent in the radiated frequency band and also acts as radome to protect the sensitive antenna structures.

To visualize the current operation step, the radar provides four LEDs:

- *POWER* indicates that the module is connected to power supply and is ready for operation,
- *DATA* indicates that a communication process between radar and host is in progress (e.g. transmission of user commands or measurement data of any type),
- *TARGET* indicates that a target object has been detected by a high-level algorithm,
- *AUX* indicates unexpected radar behavior, e.g. in case of an internal error.

On the back side, the sentire™ Radar Module provides a multi-functional cable for power supply and communication interface (see figure 2.1 C). Depending on the built-in Interface board, the radar supports one of three bus systems: SPI, CAN or Ethernet (for more details please refer to sections 2.2 to 2.4).

2.2 SPI Interface

A sentire™ Radar Module equipped with an SPI interface can be generally accessed via two different methods:

1. direct SPI access using the connection pins described in section 2.2.1,
2. using the provided SPI-to-USB converter (described in section 2.2.3) and the corresponding FTDI computer library.

2.2.1 SPI Pin Description

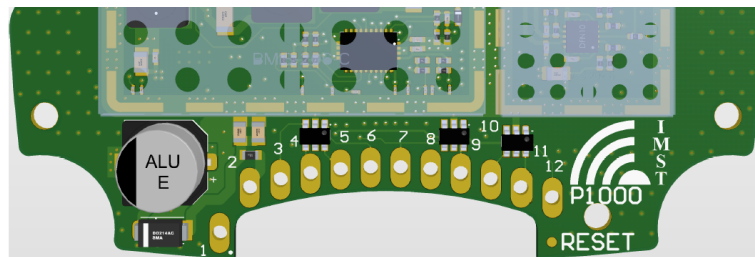


Figure 2.2: Connection pins of the SPI interface board

Figure 2.2 shows the connection pins of the SPI interface board. The pins which are relevant for DC power supply and for the SPI interface are listed and described in table 2.1. All other pins (3 to 7) serve for development and debugging reasons and should not be connected to any external voltage.

Pin	Name	min. [V]	max. [V]	Description
1	GND	—	—	Ground for power supply
2	VCC	10.5	13.0	Power supply voltage
8	SPI_LVL	1.65	5.5	SPI Logic Level (INPUT)
9	SPI_CLK	-0.5	5.5	SPI Clock Frequency (INPUT)
10	SPI_MOSI	-0.5	5.5	SPI Master Data Out (INPUT)
11	SPI_MISO	-0.5	5.5	SPI Master Data In (OUTPUT)
12	SPI_CS	-0.5	5.5	SPI Chip Select (INPUT)

Table 2.1: Pin description of the SPI interface board

Parameter	Value
SPI Mode	Master
SPI Clock Rate	0.1 – 2 MHz
SPI Chip Select	low active
SPI Master Transmit Data	on falling edge
SPI Master Receive Data	on rising edge
SPI Word Length (read & write)	16 bits
SPI Byte Order (read & write)	LSB first

Table 2.2: SPI configuration parameters for the controlling system

2.2.2 SPI Parameters

The SPI interface of the sentireTM Radar Module is directly connected to the DSC and operates in slave mode (i.e. the controlling system is SPI master). The outside TTL level of the SPI interface pins can be configured using the pin SPI_LVL (see table 2.1) in order to enable a two-directional translation between any of the 1.8 V, 2.5 V, 3.3 V or 5.5 V voltage nodes. Table 2.2 shows a list of parameters required for configuration of the controlling system.

2.2.3 SPI-to-USB Converter

The SPI version of the sentireTM Radar Module is optionally delivered with the TTL-to-USB Serial Converter Cable C232HM-DDHSL-0 which is based on the FTDI chip FT232H (for more details please refer to www.ftdichip.com). In the present application, the FTDI chip is configured to act as SPI-to-USB converter between the Radar Module and the controlling computer. If connected to the USB port of a computer, the FTDI chip creates a virtual COM port (see figure 2.3) which can be accessed using FTDI's driver interface FTD2XX.dll. Please note, that the FTDI driver must be installed in advance before using the USB port.

Additionally, IMST provides a Python user interface package which is able to communicate

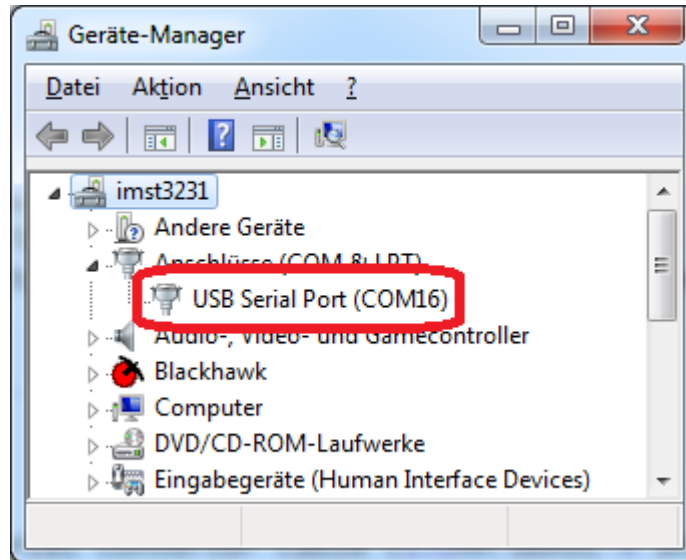


Figure 2.3: Virtual COM port appearing in the Windows Device Manager after connecting the TTL-to-USB Serial Converter Cable

with the Radar Module through the FTDI library (see /CMD/). The software IMST SenTool (available as part of the Development Kit) supports this communication interface. In order to allow the user a quick start with programming the Radar Module, a simple Python example code is also provided.

2.3 CAN Bus Interface

Similarly to the SPI version, a sentireTM Radar Module equipped with a CAN interface can be also be accessed in two general ways:

1. direct CAN access using the connection pins described in section 2.3.1,
2. using an optional CAN-to-USB converter (described in section 2.3.3).

2.3.1 CAN Pin Description

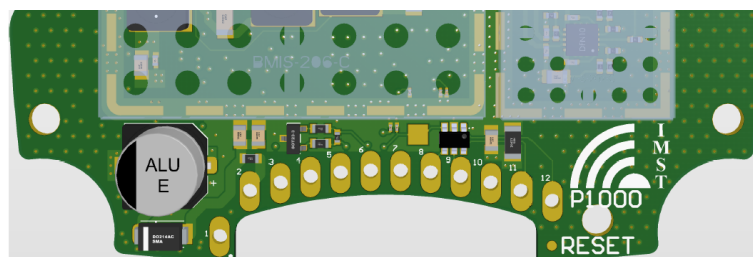


Figure 2.4: Connection pins of the CAN interface board

Pin	Name	min. [V]	max. [V]	Description
1	GND	—	—	Ground for power supply
2	VCC	10.5	13.0	Power supply voltage
5	CANH	-2.0	7.0	High level CAN bus line
6	CANL	-2.0	7.0	Low level CAN bus line
3, 4, 7, 10	GND_S	—	—	Shielding ground

Table 2.3: Pin description of the CAN interface board

Figure 2.4 shows the connection pins of a CAN bus version of the Radar Module, table 2.3 provides a description of the corresponding functionality. The DC power supply pins 1 and 2 are similar to the SPI version, while the pins 5 and 6 provide a two-wire CAN bus interface corresponding to ISO 11898-2 (Highspeed-CAN standard). The use of connection pins indicated as GND_S is optional (for long range transmission, these pins can be connected to the cable shielding). All other pins serve for development and debugging reasons and should not be connected to any external voltage.

Internally, the CAN bus data lines are terminated using a $120\ \Omega$ resistor. Depending on the final CAN network configuration, it may be necessary to remove the resistors from some modules in order to ensure an appropriate network impedance.

2.3.2 CAN Parameters

Parameter	Value
Baud Rate [kbaud]	10, 20, 50, 100, 125, 250, 500, 800, 1000
Default Baud Rate [kbaud]	1000
CAN Node ID	0 to 62 (default: 2)
Broadcast CAN Node ID	63
CAN Word Length (read & write)	16 bits
CAN Byte Order (read & write)	LSB first

Table 2.4: CAN configuration parameters of the Radar Module

When connected via CAN bus interface the sentireTM Radar Module can be addressed using its node ID (which should be unique in the CAN network) as well as a broadcast ID which is identical for all radar modules. After a successful connection, the CAN bus parameters of a radar module can be accessed and optionally modified as described in the Communication Interface manual (see /CMD/). Table 2.4 summarizes the CAN bus configuration parameters.

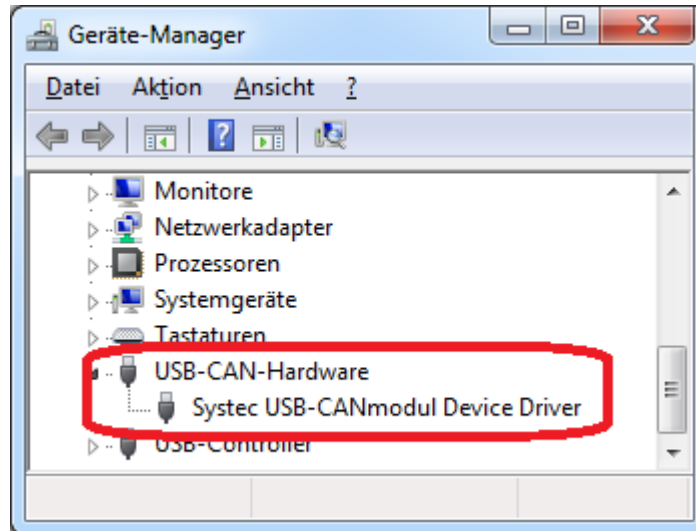


Figure 2.5: USB-CAN device appearing in the Windows Device Manager after connecting the Systec CAN-to-USB Converter

2.3.3 CAN-to-USB Converter

In order to connect a sentireTM Radar Module to a computer, an optional CAN-to-USB converter can be applied. IMST recommends to use Systec's compact USB-CANmodul1 which acts as translator between the USB port and the CAN network containing one or several radar sensors (for more details please refer to www.systec-electronic.com). If connected to the USB port of a computer, the USB-CANmodul1 appears in the Windows Device Manager as shown in figure 2.5. Please note, that the corresponding driver must be installed in advance before using the USB port.

Additionally, IMST provides a Python user interface package which is able to communicate with the Radar Module through the presented CAN-to-USB converter (see /CMD/). The software IMST SenTool (available as part of the Development Kit) supports this communication interface. In order to allow the user a quick start with programming a CAN bus Radar Module, a simple Python example code is also provided.

2.4 Ethernet Interface

The sentireTM Radar Module can be connected to a LAN using its Ethernet interface. Even if the Radar Module in the most cases will be delivered with an attached cable, section 2.4.1 contains a pin description of the Ethernet Interface board. The corresponding communication parameters are listed in section 2.4.2.

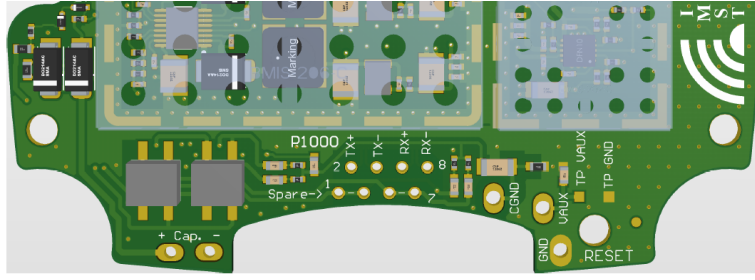


Figure 2.6: Connection pins of the Ethernet interface board

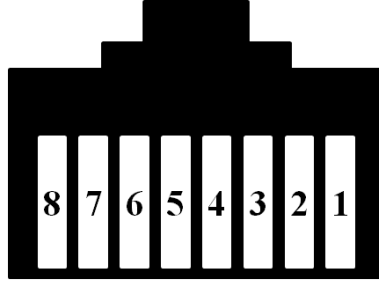


Figure 2.7: Pin numbering of a RJ-45 Jack (frontview)

2.4.1 Ethernet Pin Description

The Ethernet version of the Radar Module is powered over the data lines or over the unused data lines of the 8-wire Ethernet cable and is compliant to the IEEE 802.3af PoE standard. Table 2.5 lists the relation between the pins of a RJ-45 Jack with a T568B pinout and the pins of the interface board. Since the IEEE 802.3af standard defines several possibilities to connect the DC power supply voltage to the Ethernet data lines, different modes are mentioned in Table 2.5. One of the following defined modes for the pins of the interface board can be chosen, where mode B1 is the standard mode:

- pins 2+4 $\rightarrow$ GND, pins 6+8 $\rightarrow$ 48 V (PoE Mode A1),
- pins 6+8 $\rightarrow$ GND, pins 2+4 $\rightarrow$ 48 V (PoE Mode A2),
- pins 1+3 $\rightarrow$ GND, pins 5+7 $\rightarrow$ 48 V (PoE Mode B1),
- pins 5+7 $\rightarrow$ GND, pins 1+3 $\rightarrow$ 48 V (PoE Mode B2).

Additionally to the interface data lines, the cable shielding should be connected to the pin CGND on the interface board.

2.4.2 Ethernet Parameters

When connected via Ethernet interface, the radar can be addressed using its Ethernet configuration parameters (see table 2.6). After a successful connection, the communication parameters of a radar module can be accessed and optionally modified as described in the

Pin (Jack)	Pin (Interface Board)	Name	Colour Coding	Description
1	2	Tx+	 orange striped	Tx+ data line, 48 V/GND (PoE Modes A2/A1)
2	4	Tx-	 orange	Tx- data line, 48 V/GND (PoE Modes A2/A1)
3	6	Rx+	 green striped	Rx+ data line, GND/48 V (PoE Modes A2/A1)
4	3	Spare	 blue	48 V/GND (PoE Modes B2/B1)
5	1	Spare	 blue striped	48 V/GND (PoE Modes B2/B1)
6	8	Rx-	 green	Rx- data line, GND/48 V (PoE Modes A2/A1)
7	5	Spare	 brown striped	GND/48 V (PoE Modes B2/B1)
8	7	Spare	 brown	GND/48 V (PoE Modes B2/B1)

Table 2.5: Pin relation between RJ-45 jack and the interface board with T568B convention (graphics from Wikipedia)

Parameter	Value
Default Port	1024
Default IP Address	192.168.0.2
Gateway	192.168.0.1
Subnet Mask	255.255.0.0

Table 2.6: Ethernet configuration parameters of the Radar Module

Communication Interface manual (see /CMD/). One host can be connected to several radar-modules, when they have different connection parameters (port and IP address). The default connection parameters are valid 10 seconds after power-up. After this time the Radar Module is reconnected with the user-defined connection parameters.

Additionally, IMST provides a Python user interface package which is able to communicate with the Radar Module through the Ethernet interface (see /CMD/). The software IMST SenTool (available as part of the Development Kit) supports this communication interface. In order to allow the user a quick start with programming an Ethernet Radar Module, a simple Python example code is also provided.

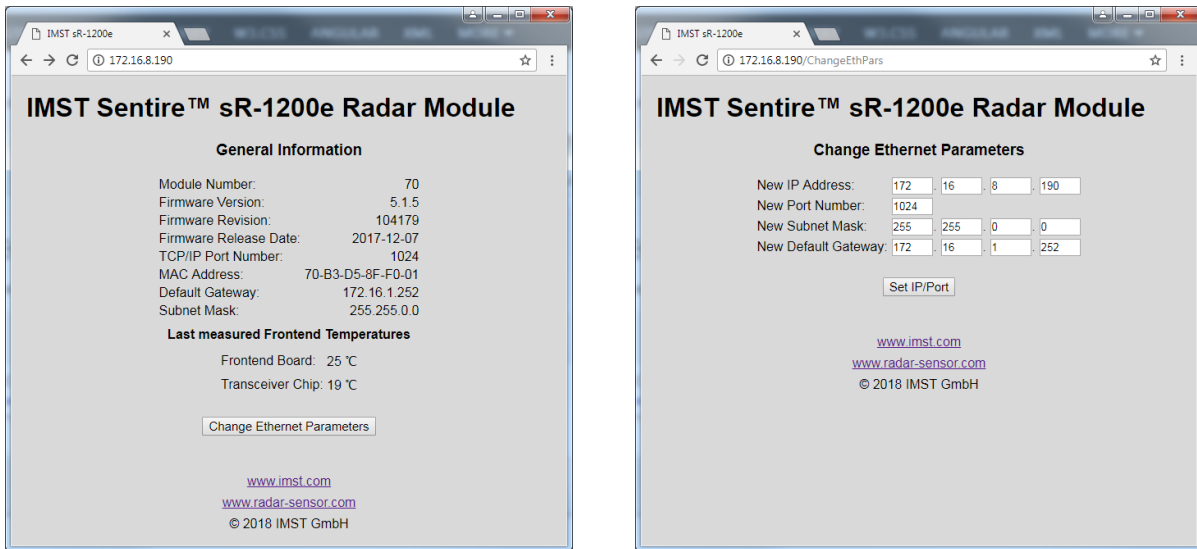


Figure 2.8: HTTP information page (a) and page to change IP address and Port (b)

2.4.3 HTTP Server

The sentireTM Radar Module with Ethernet interface also runs a small HTTP server on the default port number 80. This means the radar will answer on a request from e.g. a browser to its standard IP address. The answer is a *HTML* page with module specific information like in figure 2.8 a. When clicked on the button *Change Ethernet Parameters* a page with the possibility to change the radars IP address and Port will be shown (see figure 2.8 b). The new IP address and Port can then be set by clicking the button *Set IP/Port*. Dependent on the used browser a message will pop up which informs if the setting was successful or not. There are some additional possible request URLs for specific operation which are described in /CMD/.

Note: If the IP address of the module is unknown, it is possible to get and change the current IP and Port configuration via a browser. Therefore the host computer should be in the same subnet (192.168.0.3 - 192.168.0.255) as the default IP address of the radar. Within the first 10 seconds after a power reset of the radar, the URL "192.168.0.2/ChangeEthPars" should be called by a browser. This page will show the current saved configuration. It is also possible to change these values here within the 10 seconds limit.

2.5 DC Power Supply

The sentireTM Radar Module equipped with SPI or CAN interface is powered using an external 12V DC power adapter (provided with the sensor). The power adapter has an 2-pin connector which is *not* protected against wrong polarity. As the radar is equipped with a reverse voltage protection, wrong DC polarity will not damage the sensor, but an operation will be not possible in this case (a correct power supply is indicated by the *POWER* LED as described in section 2.1). Alternatively, the user can apply his own

DC power supply to the radar sensor, as long as it complies to the specification listed in chapter 3.

3. Technical Specifications

3.1 Datasheet

Parameter	Specification
<i>General</i>	
Modulation	CW / FMCW
Operating Frequency	24.00 GHz – 24.25 GHz (@ ISM band)
Number of Channels	1 Transmit-Channel (Tx), 2 Receive-Channels (Rx) with I/Q demodulator for each channel
Data Interface	SPI, CAN, Ethernet (with PoE)
Certification	CE Approval (Ethernet version only)
<i>Antenna</i>	
Antenna Type	Integrated Patch Antenna
Antenna Characteristics	65° Azimuth and 24° Elevation
Antenna Polarization	linear
<i>Measurement</i>	
Measurement Range	0.6 m – 307 m (@ ISM band)
Range Resolution	max. 0.6 m (@ ISM band)
Measurement Speed	±3200 m/s (theoretical value @ 24 GHz)
Speed Resolution (1 – 100 ms)	6.25 m/s – 6.25 cm/s (@ 24 GHz)
Angle Measurement Accuracy	2° – 3°
<i>Operation Parameters</i>	
Frequency Ramp Duration	1 ms – 100 ms
Update Rate	typ. 10 Hz – 200 Hz (see section 3.2)
EIRP Output Power	typ. 10 dBm – 19 dBm (tunable)
Operating Temperature	min. -40 °C, max. 60 °C (see section 3.3)
<i>Power Supply</i>	
Operation Voltage	10.5 V – 13 V, 44 V – 54 V PoE
Standby Power	3.0 W
Operating Power	4.5 W

Parameter	Specification
<i>Housing</i>	
Dimension (L × W × H)	98.0 mm × 87.0 mm × 42.5 mm (Housing) 114.0 mm × 87.0 mm × 42.5 mm (with Bushing)
Weight	280 g
Mounting	4 Mounting Holes (Ø 5 mm)
Cable and Connector	Shielded 9 Wire Cable (SPI/CAN) CAT-6 with RJ45 Female Plug (Ethernet)
Protection Code	IP 65

3.2 Update Rate

The maximum possible measurement update rate depends on many parameters as ramp duration, signal processing time and the amount of transmitted data samples. Therefore, this value is exemplarily given for three typical applications. In the following examples, the use of the ISM band and a maximum distance of 30 m is assumed:

- FMCW measurement and transmission of TD data from 4 channels → 15.5 Hz,
- FMCW measurement and transmission of FD data (amplitude and object angle) from 2 channels → 98 Hz,
- FMCW measurement and transmission of FD data (amplitude only) from 1 channel → 180 Hz.

3.3 Operating Temperature

The maximum possible operating temperature mainly depends on the duty cycle of the RF transmitter (please note that the duty cycle is always 100% if the *Power Save* mode is disabled). The following table shows the internal temperature of the radar module at an environment temperature of 60 °C for different duty cycles:

Duty Cycle	Radar Temperature
1%	81 °C
10%	82 °C
25%	83 °C
50%	84 °C
100%	86 °C (→ overheat)

4. Radar Fundamentals

The following chapter provides a short overview about the basics of radar technology employed in the IMST's sentireTM Radar Module. For more details and applications the reader is asked to refer to technical literature.

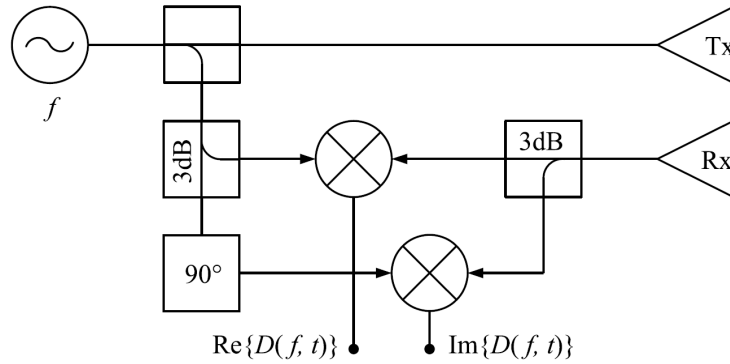


Figure 4.1: Generalized block diagram of a bistatic CW/FMCW radar system with one transmitter and one receiver channels and I/Q demodulator

4.1 Principle of Operation

The operation principle of a typical CW/FMCW radar system is demonstrated in figure 4.1 and will be described below. A radar consists of the following components:

- Signal source for generation of a (frequency-modulated) signal,
- Transmitter antenna (Tx),
- One or several receiver antennas (Rx),
- One or several mixers for generation of the Intermediate Frequency (IF),
- Digital sampling unit (A/D converter),
- Couplers, filters, amplifiers, phase shifters (these components are ignored here).

Assuming that the Tx antenna transmits a signal with the frequency f_T and the Rx antenna at the same time receives the frequency f_R , the resulting intermediate frequency f_I can be calculated using the formula:

$$f_I = f_T - f_R. \quad (4.1)$$

It is obvious that the value of f_I can be positive as well as negative (depending on f_T and f_R), which explains the need for the I/Q demodulator in figure 4.1. Nevertheless, the following calculation will be based on real-valued sampling. Now we assume that the IF signal is digitized with a sampling frequency f_S for the time interval T . This yields a discrete Time-Domain (TD) signal with

$$N_S = T \cdot f_S \quad (4.2)$$

real data samples, which can be transformed to Frequency-Domain (FD) using a Fast Fourier Transform (FFT). The result is a complex-valued signal consisting of $N_S/2$ data samples with a frequency step of

$$\delta_f = \frac{1}{T}. \quad (4.3)$$

This expression represents the system's frequency resolution and allows to assign a data sample on position n (also called "bin number n " with $n \geq 0$) of the IF signal spectrum to a certain intermediate frequency:

$$f_I(n) = \delta_f \cdot n. \quad (4.4)$$

The maximum intermediate frequency f_{max} which can unambiguously be measured by the system, is calculated using the equations 4.2 to 4.4:

$$f_{max} = \delta_f \cdot \frac{N_S}{2} = \frac{N_S}{2 \cdot T} = \frac{T \cdot f_S}{2 \cdot T} = \frac{f_S}{2}. \quad (4.5)$$

Equation 4.5 simply states that the sampling frequency f_S must be twice as large as highest frequency f_{max} included in the IF signal. This is nothing else than the representation of the Nyquist-Shannon sampling theorem. In the IMST's sentireTM Radar Module, the TD signal always consists of 1024 data samples. Hence, the corresponding maximum frequency is given by the equation:

$$f_{max} = \frac{512}{T}. \quad (4.6)$$

4.2 FMCW Mode

If operating in FMCW mode, the radar generates a linear frequency ramp of the duration T covering the bandwidth B (see figure 4.2). Assuming a point scatterer at a distance R , the propagation time from transmitter to the receiver can be expressed as:

$$T_0 = \frac{2 \cdot R}{c_0} \quad (4.7)$$

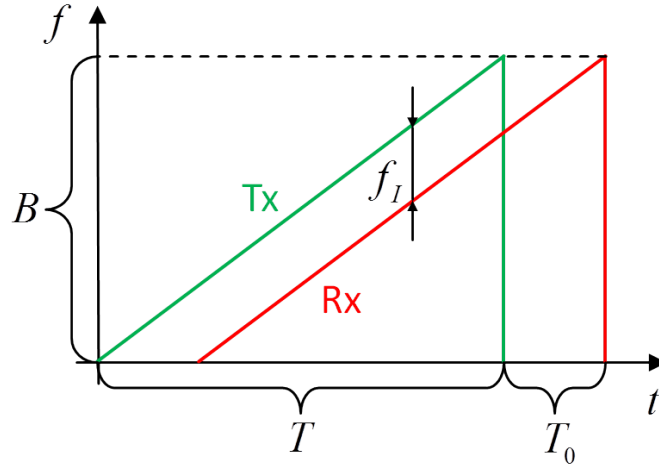


Figure 4.2: Operation principle of an FMCW radar

with c_0 representing the speed of light. Thus, the signal arriving at the receiver antenna (Rx) is delayed by the time T_0 compared with the signal at the transmitter antenna (Tx). Consequently, the frequencies f_T and f_R appearing at the mixer inputs are also slightly different. This difference exactly corresponds to the intermediate frequency f_I (compare to equation 4.1) and can easily be calculated using figure 4.2:

$$f_I = \frac{B \cdot T_0}{T} = \frac{2 \cdot B \cdot R}{T \cdot c_0}. \quad (4.8)$$

This gives us the possibility to assign any target distance R to a fixed intermediate frequency f_I :

$$R(f_I) = \frac{T \cdot c_0}{2 \cdot B} \cdot f_I. \quad (4.9)$$

The corresponding range resolution δ_r of the radar can be derived from equation 4.9 using the frequency resolution δ_f known from equation 4.3:

$$\delta_r = R(\delta_f) = \frac{T \cdot c_0}{2 \cdot B} \cdot \delta_f = \frac{c_0}{2 \cdot B}, \quad (4.10)$$

which is the well-known formula valid for *all* radar sensors. Analyzing FD data from an FMCW measurement, it is also possible to say that the “range bin number n ” corresponds to a target distance of:

$$R(n) = \delta_r \cdot n. \quad (4.11)$$

The maximum range which can (theoretically) be measured by the radar, is calculated analogue to equation 4.10 using the maximum frequency f_{max} from equation 4.5:

$$R_{max} = R(f_{max}) = \frac{T \cdot c_0}{2 \cdot B} \cdot f_{max} = \frac{c_0 \cdot N_S}{4 \cdot B}. \quad (4.12)$$

For the special case of the IMST Radar Module ($N_S = 1024$), the maximum measurement range corresponds to

$$R_{max} = \frac{c_0 \cdot 256}{B}. \quad (4.13)$$

4.3 Doppler Mode

If operating in CW (or Doppler) mode, the radar transmits a single frequency f_0 without any modulation, i.e. with a bandwidth of $B = 0$. From equation 4.10 it becomes clear, that no range resolution is possible in this case, as the transmitted and received frequencies (f_T and f_R) are equal for all *static* targets. Anyway, this behavior significantly changes for *moving* objects: each radial velocity v between radar and target causes a frequency shift of the reflected signal, the so called Doppler frequency

$$f_D = \frac{2 \cdot f_0 \cdot v}{c_0}. \quad (4.14)$$

For the sake of completeness, it should be mentioned that equation 4.14 is only valid for non-relativistic cases ($|v| \ll c_0$), which is usually fulfilled for the most realistic applications. For a target moving *away* from sensor (v is defined as positive), we expect to receive a signal with the frequency

$$f_R = f_T - f_D, \quad (4.15)$$

leading to an intermediate frequency of:

$$f_I = f_T - f_R = f_D = \frac{2 \cdot f_0 \cdot v}{c_0}. \quad (4.16)$$

Again, we can establish a link between a certain target velocity v and the corresponding intermediate frequency f_I :

$$v(f_I) = \frac{c_0}{2 \cdot f_0} \cdot f_I. \quad (4.17)$$

Analogue to equations 4.10 and 4.12, the velocity resolution δ_v as well as the maximum velocity v_{max} can be calculated by inserting δ_f and f_{max} into equation 4.17:

$$\delta_v = v(\delta_f) = \frac{c_0}{2 \cdot f_0} \cdot \delta_f = \frac{c_0}{2 \cdot f_0 \cdot T}. \quad (4.18)$$

$$v_{max} = v(f_{max}) = \frac{c_0}{2 \cdot f_0} \cdot f_{max} = \frac{c_0 \cdot N_S}{4 \cdot f_0 \cdot T}. \quad (4.19)$$

For the special case of the IMST Radar Module ($N_S = 1024$), the maximum unambiguously measured target velocity corresponds to

$$v_{max} = \frac{c_0 \cdot 256}{f_0 \cdot T}. \quad (4.20)$$

In comparison to FMCW measurements, an additional difficulty appear in the CW (or Doppler) mode. Due to equation 4.18, it is possible to say that the “velocity bin number n ” corresponds to a radial target velocity of

$$v(n) = \delta_v \cdot n. \quad (4.21)$$

Unfortunately, this statement doesn't take into account the direction of the target motion: positive as well as negative velocities of the same magnitude appear at an identical position in the IF signal spectrum. To resolve this ambiguity, the I/Q demodulator of the radar sensor (see figure 4.1) can be used. Positive and negative Doppler frequencies can easily be separated by using complex instead of real-valued IF signals. The mathematical background of this operation can be extracted from Fourier Transforms theory.

4.4 Range-Doppler-Ambiguity

In sections 4.2 and 4.3, the target distance R as well as its radial velocity v have been linked to the intermediate frequency f_I measured by the radar sensor (compare equations 4.8 and 4.16). If dealing with moving targets in FMCW operation mode, this leads to an effect called Range-Doppler-Ambiguity. The main problem is the “mixing” of two independent parameters (R and v) to a single measurement value (f_I), resulting in erroneous range measurements.

The expected range error appearing at a target moving with a radial velocity v can be estimated by superposition of equations 4.8 and 4.16:

$$\Delta R(v) = \frac{f_0 \cdot T}{B} \cdot v. \quad (4.22)$$

In case of spatially well separated targets, this limitation can be overcome by alternating transmission of rising (Up) and falling (Down) frequency ramps. If the same target can be localized in both data sets, its parameters R and v can unambiguously be reconstructed by measuring the corresponding position in the Up and Down signal spectra. In case of noisy data with many spatially close targets, a unique reconstruction of target parameters is usually not possible.

4.5 Angle Measurement

The IMST's sentireTM Radar Module is equipped with two independent receiver antennas for measuring the Angle-of-Arrival of a received signal. This feature can be used to generate a 2-dimensional reflectivity map of the illuminated scene. To derive the formula for the view angle extraction, two receiver antennas separated by a baseline b are assumed (see figure 4.3). Further, a plane phase front shall arrive from the direction defined by the angle α . Consequently, the incident wave arrives at antenna Rx1 a little bit later compared with antenna Rx2. With the center frequency f_c , this causes a phase shift φ between both receiver antennas, given by the formula:

$$\varphi = \frac{2\pi \cdot f_c \cdot d}{c_0}. \quad (4.23)$$

The parameter d describes the additional distance the wave must travel to reach the antenna Rx1. This value is defined by

$$d = b \cdot \sin(\alpha), \quad (4.24)$$

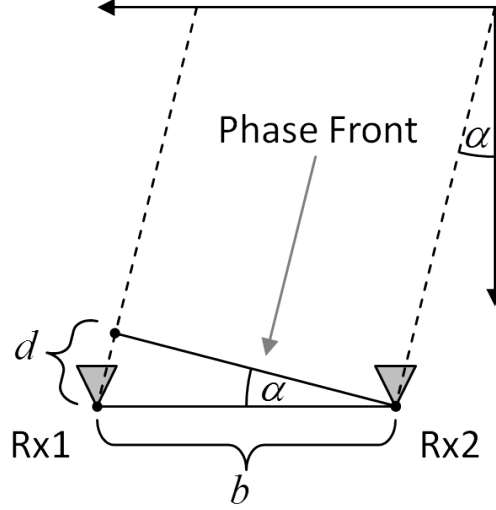


Figure 4.3: Estimation of the View Angle using two receiver antennas

so we can expect a phase difference of

$$\varphi = \frac{2\pi \cdot f_c \cdot b \cdot \sin(\alpha)}{c_0} \quad (4.25)$$

between the signals received by the antennas Rx1 and Rx2. Equation 4.25 can be inverted to derive the final relation between the measured phase difference φ and the angle α :

$$\alpha(\varphi) = \arcsin\left(\frac{\varphi \cdot c_0}{2\pi \cdot f_c \cdot b}\right). \quad (4.26)$$

Finally, some physical limitations must be added to equation 4.26:

- the assumption of a plane phase front (see figure 4.3) is only valid for targets located in the far-field of the antenna,
- to avoid ambiguities, the argument of the $\arcsin()$ function must be in limits $[-1, 1]$,
- as the baseline b is defined for lumped antenna elements, the formula gets imprecise for large angles α ,
- non-ideal hardware components, as transmission lines, sensor housing, radome or PCB ground planes, additionally influence the antenna behavior as well as the estimated view angles.

In the special case of the IMST Radar Module, a baseline of 6.25 mm has been selected, corresponding to a half wave length at 24 GHz.

Appendix

A Referenced Documents

The following table contains a list of documents referenced by this manual.

Ref. ID	Document Name	Source
/RAD/	sR1200_Radar_Manual.pdf	IMST GmbH
/CMD/	sR1200_Command_Interface.pdf	IMST GmbH
/GUI/	sR1200_Development_Kit.pdf	IMST GmbH

B List of Abbreviations

ASCII:	American Standard Code for Information Interchange
CAN:	Controller Area Network
CE:	European Community
CW:	Continuous Wave
DC:	Direct Current
DSC:	Digital Signal Controller
EEPROM:	Electrically Erasable Programmable Read-Only Memory
EIRP:	Equivalent Isotropically Radiated Power
ELV:	Extra-Low Voltage
ESD:	Electrostatic Discharge
FD:	Frequency Domain
FFT:	Fast Fourier Transform
FMCW:	Frequency Modulated Continuous Wave

FTDI:	Future Technology Devices International Ltd. (company name)
GUI:	Graphical User Interface
HDD:	Hard Disk Drive
HT:	Human Tracker (algorithm)
ID:	Identifier
IEC:	International Electrotechnical Commission
IEEE:	Institute of Electrical and Electronics Engineers
IF:	Intermediate Frequency
IP:	Internet Protocol
ISM:	Industrial, Scientific and Medical (frequency band)
JPEG:	Joint Photographic Experts Group (file format)
LAN:	Local Area Network
LED:	Light-Emitting Diode
MAC:	Media Access Control
MB:	Megabyte (corresponds to 1.000.000 bytes)
PoE:	Power over Ethernet
RCS:	Radar-Cross-Section
RF:	Radio Frequency or Range Finder (algorithm)
Rx:	Receiver (antenna, etc.)
SPI:	Serial Peripheral Interface
TD:	Time Domain
Tx:	Transmitter (antenna, etc.)
USB:	Universal Serial Bus

C List of Figures

2.1	IMST sentire TM Radar Module (A: radome and antenna position, B: LED indicators, C: interface and power supply cable)	5
2.2	Connection pins of the SPI interface board	6
2.3	Virtual COM port appearing in the Windows Device Manager after connecting the TTL-to-USB Serial Converter Cable	8
2.4	Connection pins of the CAN interface board	8
2.5	USB-CAN device appearing in the Windows Device Manager after connecting the Systec CAN-to-USB Converter	10
2.6	Connection pins of the Ethernet interface board	11
2.7	Pin numbering of a RJ-45 Jack (frontview)	11
2.8	HTTP information page (a) and page to change IP address and Port (b) .	13
4.1	Generalized block diagram of a bistatic CW/FMCW radar system with one transmitter and one receiver channels and I/Q demodulator	17
4.2	Operation principle of an FMCW radar	19
4.3	Estimation of the View Angle using two receiver antennas	22

D List of Tables

2.1	Pin description of the SPI interface board	7
2.2	SPI configuration parameters for the controlling system	7
2.3	Pin description of the CAN interface board	9
2.4	CAN configuration parameters of the Radar Module	9
2.5	Pin relation between RJ-45 jack and the interface board with T568B convention (graphics from Wikipedia)	12
2.6	Ethernet configuration parameters of the Radar Module	12

Important Notices

Regulatory Compliance Information

The use of radio frequencies is limited by national regulations. The Radar Module has been designed to comply with the European Union's R&TTE (Radio & Telecommunications Terminal Equipment) directive 1999/5/EC and can be used license free within the European Union at 24–24.25 GHz. Nevertheless, please refer to your national frequency regulation if there is any need for a notification at least.

The Radar Module (with housing) has been designed to be used with suitable IT-equipment to collect, store and process the radar data (referred as “final system”). According to the R&TTE directive, the declaration of compliance with the essential requirements of the R&TTE directive is within the responsibility of the manufacturer of the final system. A declaration of conformity (DoC) for the Radar Module is available from IMST GmbH on request.

The applicable regulation requirements are subject to change. IMST GmbH does not take any responsibility for the correctness and accuracy of the aforementioned information. National laws and regulations, as well as their interpretation can vary with the country. In case of uncertainty, it is recommended to contact either IMST's accredited Test Center or to consult the local regulatory authority of the respective country of use.

Disclaimer

IMST GmbH points out that all information in this document is given on an “as is” basis. No guarantee, neither explicit nor implicit is given for the correctness at the time of publication. IMST GmbH reserves all rights to make corrections, modifications, enhancements, and other changes to its products and services at any time and to discontinue any product or service without prior notice. It is recommended for customers to refer to the latest relevant information before placing orders and to verify that such information is current and complete. All products sold and delivered are subject to “General Terms and Conditions” of IMST GmbH, supplied at the time of order acknowledgment.

IMST GmbH assumes no liability for the use of its products and does not grant any licenses for its patent rights or for any other of its intellectual property rights or third-party rights. It is the customer's duty to bear responsibility for compliance of systems or units in which

products from IMST GmbH are integrated with applicable legal regulations. Customers should provide adequate design and operating safeguards to minimize the risks associated with customer products and applications. The products are not approved for use in life supporting systems or other systems whose malfunction could result in personal injury to the user. Customers using the products within such applications do so at their own risk.

Any reproduction of information in datasheets of IMST GmbH is permissible only if reproduction is without alteration and is accompanied by all given associated warranties, conditions, limitations, and notices. Any resale of IMST GmbH products or services with statements different from or beyond the parameters stated by IMST GmbH for that product/solution or service is not allowed and voids all express and any implied warranties. The limitations on liability in favour of IMST GmbH shall also affect its employees, executive personnel and bodies in the same way. IMST GmbH is not responsible or liable for any such wrong statements.

Copyright © 2016, IMST GmbH

Contact Information

IMST GmbH
Carl-Friedrich-Gauss-Str. 2-4
47475 Kamp-Lintfort
Germany
Phone: +49 2842 981 0
Fax: +49 2842 981 199
E-Mail: radar@imst.de
Internet: www.radar-sensor.com